Table 1: Datasets of the 1.8.0 study, the 1.9.0 pilot and the planned 1.9 studies

study	status	tasks	base	logical	families	widths	devices	noise settings	budgets	r/half	designs/half
dev18	historical development (1.8.0)	88	44	44	8	3–6	3	native 88	10: 1×10^3–3×10^7	60	6
conf18	historical confirmation (1.8.0)	58	29	29	8	3–6	3	native 58	10: 1×10^3–3×10^7	60	6
pilot19	exploratory pilot (1.9.0)	54	27	27	5	3–5	3	g2=2 10, native 18, ro=2 4, s=0.5 5, s=2 9, t=2 8	4: 1×10^3–3×10^7	24	3
fulldev19	planned development	246	82	82	8	3–6	3	chi=0.25 16, g2=2 34, native 80, ro=2 23, s=0.5 21, s=2 38, t=2 34	10: 1×10^3–3×10^7	60	6
conf19	planned confirmation	64	32	32	5	3–6	3	chi=0.25 5, g2=2 5, native 28, ro=2 5, s=0.5 4, s=2 9, t=2 8	10: 1×10^3–3×10^7	60	6

Counts from each configuration's expansion (historical task identities unchanged); a task is a (circuit, input, observable, layout, device) variant; base and logical circuits are the independent units. Noise settings are calibration-supplied snapshot modifiers (s: overall scale, g2: two-qubit errors, t: relaxation, ro: readout, chi: coherent fraction). Planned studies are not executed.

Table 2: The 17 candidate protocols: applicability and resources

candidate	family	learning	prerequisites	resources and shot ledger	on failure
RAW	raw	no	none	1 target circuit; B = N	none needed
RO	raw	no	2 calibration circuits ($\|0\ldots0\rangle$, X on every qubit)	1 target + 2 calibration circuits; B = c·s (calibration) + N (target); s from the supplied model's calibration-error coefficient	a singular calibration draw → no readout correction in that trial (cal_fallback)
ZNE-L	zne	no	calibration	2 folded target circuits + 2 calibration circuits; B = c·s (calibration) + N (target); s from the supplied model's calibration-error coefficient; N split over folds $\propto \lvert\eta_j\rvert\sigma_j$	cal_fallback
ZNE-R	zne	no	calibration	3 folded target circuits + 2 calibration circuits; B = c·s (calibration) + N (target); s from the supplied model's calibration-error coefficient; N split over folds	cal_fallback
ZNE-E	zne	no	calibration	3 folded target circuits + 2 calibration circuits; B = c·s (calibration) + N (target); s from the supplied model's calibration-error coefficient; N split over folds	exponential fit failure → linear fit of the same data (fit_fallback); cal_fallback
PEC	pec	no	the snapshot's median error rates (a calibration record)	one sampled circuit per shot (instances drawn from the quasi-probability), 2 calibration circuits; B = c·s (calibration) + N (target); s from the supplied model's calibration-error coefficient	cal_fallback; noninvertible model → method error (recorded)
CDR	learning	yes	non-Clifford gates; training circuits whose ideal values vary (classically computed)	16 training circuits, 1 target, 2 calibration circuits; B = 25 % training (equal per circuit) + calibration + target	degenerate training data → readout-corrected target value (learn_fallback)
NIL	nil	yes	non-Clifford gates; varying training labels	16 training circuits × k neighbors, k target neighbors, 2 calibration circuits; B = 50 % training + calibration + target (equal split)	degenerate training data → readout-corrected target value (learn_fallback)
RO-TREX	raw	no	1 twirled calibration circuit	1 target circuit with a random X layer before measurement per shot, 1 calibration circuit; B = s (calibration) + N	singular symmetric calibration → no correction (cal_fallback)
RO-TW	raw	no	a Clifford two-qubit gate set (Pauli frame compensation)	1 target circuit re-randomized per shot (N random compilations), 2 calibration circuits; B = c·s (calibration) + N (target); s from the supplied model's calibration-error coefficient	cal_fallback
RO-DD	raw	no	idle windows long enough for the sequence	1 target circuit with DD pulses, 2 calibration circuits; B = c·s (calibration) + N (target); s from the supplied model's calibration-error coefficient	cal_fallback
ZNE-L-TW	zne	no	calibration	2 folded circuits re-randomized per shot, 2 calibration circuits; B = c·s (calibration) + N (target); s from the supplied model's calibration-error coefficient; N split over folds	cal_fallback
ZNE-E-B	zne	no	the observable's eigenvalue range (known)	as ZNE-E; as ZNE-E	as ZNE-E; then projection (deterministic, never increases a trial's error)
PEC-TW	pec	no	median error rates	one sampled, twirled circuit per shot; 2 calibration circuits; B = c·s (calibration) + N (tar-	cal_fallback

Table 3: Once-only confirmation of the 1.8.0 policy by budget (full deployed policy - SBS-B)

bundle	budget	Δ [95% CI]	10^{Δ}	MSE policy	MSE SBS-B	circuits
primary: frozen configuration, simulator features (allocation-level preflight)	1×10^3	-0.127 $[-0.179, -0.082]$	0.75	1.83×10^{-3}	2.61×10^{-3}	29
	3×10^3	-0.094 $[-0.163, -0.037]$	0.81	1.28×10^{-3}	1.65×10^{-3}	29
	1×10^4	-0.134 $[-0.260, -0.027]$	0.73	6.35×10^{-4}	1.27×10^{-3}	29
	3×10^4	-0.187 $[-0.427, -0.005]$	0.65	5.39×10^{-4}	1.19×10^{-3}	29
	1×10^5	-0.240 $[-0.519, -0.043]$	0.57	4.84×10^{-4}	1.14×10^{-3}	29
	3×10^5	-0.219 $[-0.297, -0.140]$	0.60	1.60×10^{-4}	1.69×10^{-4}	29
	1×10^6	-0.119 $[-0.220, -0.034]$	0.76	1.54×10^{-4}	1.58×10^{-4}	29
	3×10^6	-0.110 $[-0.215, -0.029]$	0.78	1.48×10^{-4}	1.52×10^{-4}	29
	1×10^7	-0.106 $[-0.234, -0.007]$	0.78	7.59×10^{-5}	1.51×10^{-4}	29
	3×10^7	-0.104 $[-0.241, -0.005]$	0.79	7.55×10^{-5}	1.50×10^{-4}	29
	all	-0.144 $[-0.231, -0.076]$	0.72	5.38×10^{-4}	8.64×10^{-4}	29
secondary: default bundle, deployable features	1×10^3	-0.107 $[-0.185, -0.035]$	0.78	2.00×10^{-3}	2.61×10^{-3}	29
	3×10^3	-0.083 $[-0.170, +0.006]$	0.83	1.14×10^{-3}	1.65×10^{-3}	29
	1×10^4	-0.047 $[-0.199, +0.091]$	0.90	7.96×10^{-4}	1.27×10^{-3}	29
	3×10^4	-0.123 $[-0.384, +0.064]$	0.75	6.11×10^{-4}	1.19×10^{-3}	29
	1×10^5	-0.179 $[-0.468, +0.014]$	0.66	5.38×10^{-4}	1.14×10^{-3}	29
	3×10^5	-0.162 $[-0.268, -0.054]$	0.69	2.15×10^{-4}	1.69×10^{-4}	29
	1×10^6	-0.077 $[-0.192, +0.032]$	0.84	1.56×10^{-4}	1.58×10^{-4}	29
	3×10^6	-0.040 $[-0.187, +0.087]$	0.91	8.03×10^{-5}	1.52×10^{-4}	29
	1×10^7	-0.023 $[-0.179, +0.113]$	0.95	7.86×10^{-5}	1.51×10^{-4}	29
	3×10^7	$+0.002$ $[-0.164, +0.139]$	1.00	7.82×10^{-5}	1.50×10^{-4}	29
	all	-0.084 $[-0.179, -0.002]$	0.82	5.70×10^{-4}	8.64×10^{-4}	29

Δ = mean over base circuits g of the mean over that circuit's rows (task variants $\times$ budgets) of log10(R_policy + 10^{-6}) - log10(R_SBS-B + 10^{-6}); R = fresh-design average risk (evaluation half); $10\hat{}\Delta$ is a factor on the geometric mean of R + 10^{-6}, not on the MSE; 95 % percentile bootstrap over base circuits (1000 resamples, seed 0). Primary contrast (D41) specified after the development results and before the confirmation results; 29 base circuits, 58 tasks, 10 budgets. Per-budget rows are descriptive (10 comparisons, no multiplicity correction).

Table 4: Executed (returned-design) risk against the fresh-design label: joint audit

policy	Δ executed	Δ fresh-design	executed - fresh	rows	circuits
primary: frozen configuration, simulator features (allocation-level preflight)	-0.248 $[-0.313, -0.178]$	-0.223 $[-0.289, -0.156]$	-0.024 $[-0.047, +0.000]$	120	8
secondary: default bundle, deployable features	-0.117 $[-0.224, +0.013]$	-0.122 $[-0.199, -0.041]$	$+0.005$ $[-0.036, +0.061]$	120	8
RAW	$+1.029$ $[+0.373, +1.777]$	$+1.040$ $[+0.397, +1.797]$	-0.011 $[-0.030, +0.005]$	120	8

Eight inspected base circuits (the 1.8.0 root audit's tasks), budgets 10^4, $3{\cdot}10^5$, 10^7; deployment roots: the API's own design seeds and four independent redraws; each returned candidate executed with its verified design and the API's allocation (600 trials per row); fresh-design labels: mean of four preparation roots. Δ: base-circuit mean of $\log10(\mathrm{R} + 10^{-6})$ minus SBS-B's (the candidate the API verifies for SBS-B, executed the same way), 95 % bootstrap over base circuits. Exploratory: not a confirmation.

Table 5: The 1.9.0 pilot (exploratory): learned selection against SBS-B by split

split	policy	Δ vs SBS-B [95% CI]	held-out rows	circuits
device-base	MODEL[hgb|deployable18]	−0.003 [−0.053, +0.053]	216	27
	MODEL[hgb|size_budget]	+0.043 [+0.005, +0.079]	216	27
	MODEL[ridge|deployable18]	+0.074 [+0.004, +0.152]	216	27
	MODEL[ridge|size_budget]	+0.051 [−0.011, +0.104]	216	27
	ALL-CANDIDATE	−0.240 [−0.316, −0.170]	216	27
	SBS	+0.012 [−0.040, +0.072]	216	27
joint	MODEL[hgb|deployable18]	+0.186 [+0.075, +0.266]	12	3
	MODEL[hgb|size_budget]	+0.044 [−0.088, +0.299]	12	3
	MODEL[ridge|deployable18]	−0.017 [−0.097, +0.130]	12	3
	MODEL[ridge|size_budget]	−0.096 [−0.134, −0.063]	12	3
	ALL-CANDIDATE	−0.071 [−0.111, −0.002]	12	3
	SBS	−0.025 [−0.128, +0.096]	12	3
noise	MODEL[hgb|deployable18]	+0.103 [+0.017, +0.202]	216	27
	MODEL[hgb|size_budget]	+0.047 [+0.014, +0.085]	216	27
	MODEL[ridge|deployable18]	+0.073 [+0.019, +0.133]	216	27
	MODEL[ridge|size_budget]	+0.074 [+0.014, +0.139]	216	27
	ALL-CANDIDATE	−0.248 [−0.332, −0.171]	216	27
	SBS	+0.011 [−0.030, +0.059]	216	27
same-distribution	MODEL[hgb|deployable18+simulated]	+0.000 [−0.054, +0.056]	216	27
	ALL-CANDIDATE	−0.232 [−0.311, −0.165]	216	27
	SBS	+0.019 [−0.027, +0.072]	216	27
portfolio	expansion_fixed_policy	−0.018 [−0.042, +0.009]		
	expansion_attainable	−0.030 [−0.052, −0.007]		
	learning_within_1.6_ridge	+0.066 [−0.004, +0.127]		
	learning_within_1.8_ridge	+0.096 [+0.026, +0.163]		
	total_ridge	+0.078 [+0.010, +0.140]		
	learning_within_1.6_hgb	+0.039 [−0.015, +0.093]		
	learning_within_1.8_hgb	+0.086 [+0.014, +0.158]		
	total_hgb	+0.068 [−0.002, +0.144]		

Out-of-fold pooled results of the pilot (27 base circuits, 54 tasks, 4 budgets, noise-diverse training). On the same-distribution split the policy is the configuration the pilot's freeze chose; the other splits evaluate a declared subset of model families and feature sets, all listed (no post-hoc best). ALL-CANDIDATE is the empirical reference (best on the selection half). Exploratory: a pilot for planning, too small for conclusions; held-out-noise and joint splits test transfer to an unseen noise setting. Δ = mean over base circuits g of the mean over that circuit's rows (task variants × budgets) of log10(R_policy + 10^{-6}) - log10(R_SBS-B + 10^{-6}).

Table 6: Measured costs: labeling per task (pilot) and the planned development study

item	value	basis
pilot labeling, width 3	39 s	median of 18 tasks (R = 24 per half, 4 budgets, 3 designs/half)
pilot labeling, width 4	51 s	median of 18 tasks (R = 24 per half, 4 budgets, 3 designs/half)
pilot labeling, width 5	67 s	median of 18 tasks (R = 24 per half, 4 budgets, 3 designs/half)
fulldev19 labeling (estimate)	9.3 CPU-h	conditional estimate from the pilot's measured rates (measured on two shared cores, so contention-inflated) for fulldev19's R = 60, 10 budgets and 6 designs/half: per-trial time per trial, preparation scaled by the number of designs (x1.86); widths extrapolated: [6]. Not a proven upper bound: the coherent (chi) tasks are unmeasured
fulldev19 labeling wall time, scenario: 7 workers	1.3 h	CPU-hours / 7: a hypothetical 8-core computer with one core left free, perfect parallel efficiency; on another computer, plan –workers auto reports its own value
fulldev19 labels on disk (estimate)	0.04 GB	measured file sizes of the pilot per trial
fulldev19 analysis, main splits (model fits)	334 611	not a time: 104 split parts, fits including inner tuning; 4.0x the pilot's 83 980 fits and 51x its fits weighted by training rows (rows: 11.4x). Per-fit cost differs by family; 10 extras not counted. Time analyze –splits same-distribution first
recommendation call, qemx-rec-1.8.0	0.65 s	median classical wall time of a call that recomputed its features (1.8.0 confirmation)
recommendation call, qemx-rec-1.8.0-deployable	0.05 s	median classical wall time of a call that recomputed its features (1.8.0 confirmation)

Labeling: wall seconds of a task's supervised child process (one numerical thread; 17 candidates, every budget, both halves) — the cost of building the study, not of a recommendation. The wall time is a declared scenario (a hypothetical 8-core computer running 7 workers with perfect parallel efficiency), not a measurement and independent of the computer that generates this table. Analysis is sized by its model fits, not in hours (not measured for fulldev19).